

REVIEW-SYMPOSIUM

Altitude or heat training to increase haemoglobin mass and endurance exercise performance in elite sport

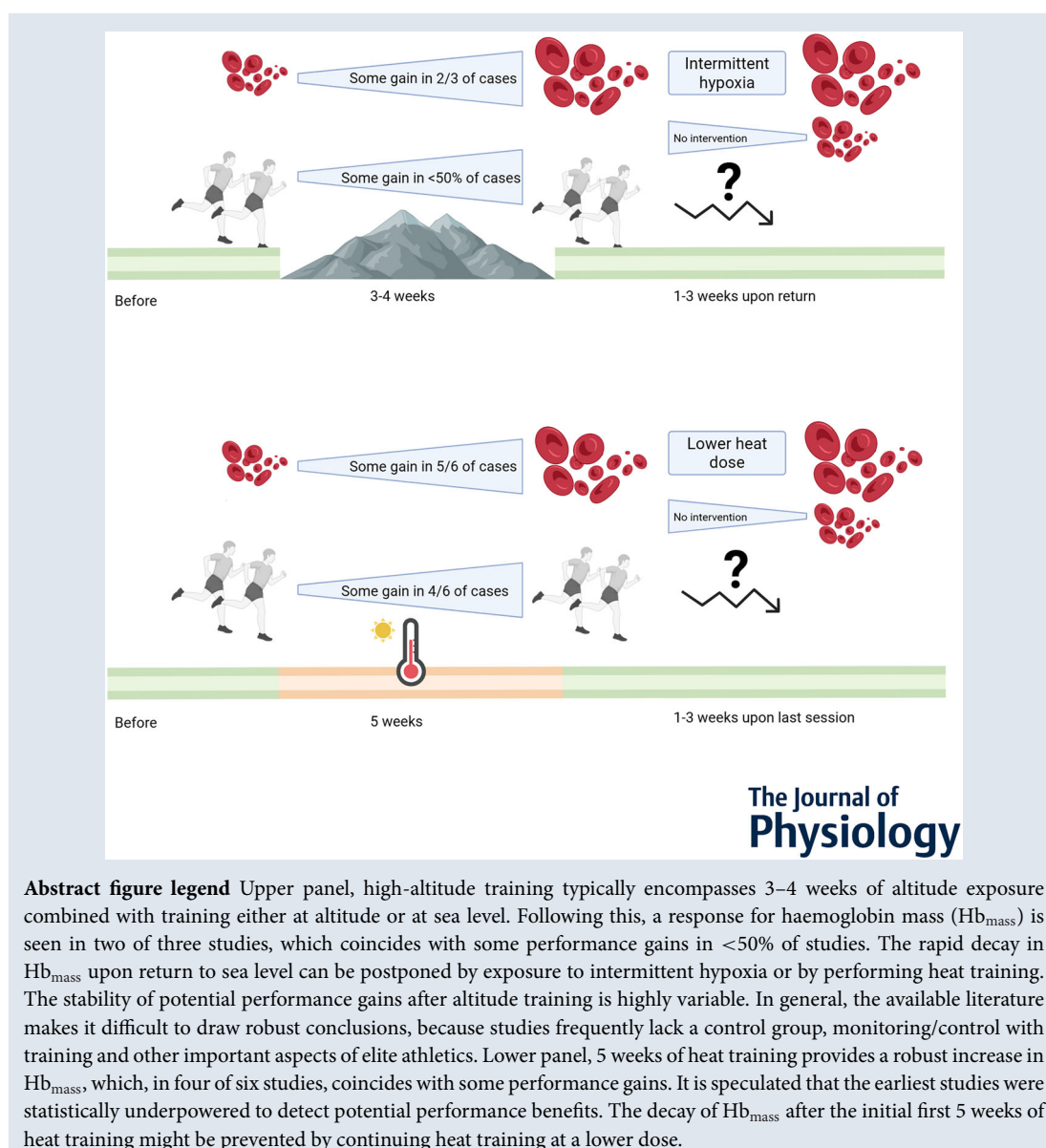
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Abstract High-altitude training has been a well-integrated tool in elite endurance sport for decades. The underlying premise is that physiological responses to high-altitude exposure, such as an increase in haemoglobin mass, are correlated with exercise performance. In this review, we highlight that physiological adaptations to a given magnitude of hypoxic exposure induce a highly variable range of responses in humans. For haemoglobin mass, this can vary from near nothing to previously unseen augmentations. At present, however, it remains inconclusive whether responses observed after one high-altitude training camp can be reproduced at the individual level during a subsequent camp. Exercise training performed in a warm environment or by wearing heat-capturing garments in a cooler environment has been termed 'heat training' and has, in recent years, become an increasingly popular alternative to altitude training. After a 3- to 5-week period of heat training, a robust increase in haemoglobin mass has repeatedly been reported, although the associated performance gains are only small. Accordingly, heat training should probably not be implemented before all other steps involved in elite training are in place and, importantly, heat training should not be performed at the expense of other important aspects of elite life. Thus, heat training might serve as a substitute for altitude training. Ultimately, it can be recommended that elite athletes experiment with either approach and that the choice of implementation should be made at the individual level, ensuring that the applied stress (hypoxia or heat) is tolerated adequately by the athlete.

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Introduction

Altitude and normobaric, hypoxic training/exposure in any of its forms has been and continues to be an important pillar for many endurance athletes/sport federations in their preparation for big sport events. In a similar manner, the implementation of heat training into the training regimes of athletes seems to have received much attention in recent years, and therefore these two strategies are the focus of this review.

It should be clear that when aiming to compete at high altitude, which for athletes might begin as low as 600 m a.s.l., or in environments with higher ambient temperatures than in one's home setting, acclimatization to those conditions, or training in similar environments while residing at home, will very probably improve exercise performance in those specific environments. The aim of this review, however, is not to discuss such an approach, but to outline the use of altitude exposure (i.e. live high–train high and live high–train

low) and heat-training strategies to augment endurance exercise performance at sea level and in typical ambient temperatures. For an update on exercise sprint training performed in hypoxia, readers are referred to a recent review by Raphaël Faiss et al. (2024).

It should also be emphasized that an exercise performance outcome might provide a 'competitive significance' long before it reaches 'statistical significance' in a scientific study. Furthermore, it should be acknowledged that it is unlikely that laboratory measures of, for example, maximal oxygen uptake ($\dot{V}_{O_{2max}}$) and similar tests, are of great relevance for evaluating real-life outdoor exercise performance, such as marathon running or bicycling races. Finally, it should also be acknowledged that in most scientific contexts, where study means are being evaluated and reported, within those means both positive and negative values are typically present. For elite exercise performance, which is highly personalized, a positive mean outcome of a study, with its included positive and negative outcomes, can perhaps at most lead

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to recommendations for individual athletes to try out a given strategy and evaluate whether it is effective for that given athlete, at that given point of time.

Altitude training

Nearly 10 years have past since our last review on altitude training (Lundby & Robach, 2016). The main conclusion stated back then was that 'Based on the available literature, the foundation to recommend altitude training to athletes is weak.' Reasons for this conclusion included that the majority of studies had been conducted without proper control groups, with inclusion of only small subject numbers and not performed in elite athletes. In addition, although markers for endurance exercise performance, such as $\dot{V}_{O_{2max}}$, haemoglobin mass (Hb_{mass}) and lactate thresholds, have typically been determined, true exercise performance was very seldomly evaluated, and reported effect sizes were typically small. A few years later and fully in agreement herewith, Bejder and Nordsborg (2018) concluded that 'Importantly, when control groups, blinding procedures, and strict scientific evaluation criteria are applied, LHTL [live high–train low] has no detectable effect on performance.' In contrast, others (Garvican et al., 2012; Wehrlin et al., 2016) have earlier expressed the opinion that negative studies might be related to the use of different methodology, although these also have been counter-argued (Lundby et al., 2012), amongst others by collaboration between both 'sides' (Keiser et al., 2017). Nonetheless, despite the pros and cons, one might ask whether the field has advanced since our last review on this topic.

Mirroring the past, a good number of studies have been published within the last decade that do not include a sea-level control group, do not control or even report exercise training volume/intensity, or any other sort of control for that matter. Although such observational analysis might provide sport federations and teams with information on whether a given altitude strategy might improve certain outcomes, it should also be clear that from a scientific point of view, such opportunistic research has only limited value. Also, it needs to be acknowledged that there is a clear difference between studies observing participation of national athletes in high-altitude training camps, where participating scientists will have only limited control over the duration, training, diet etc., and, at the other end of the spectrum, purposefully conducted scientific studies with rigorous control of external inputs, but probably not performed by top-calibre athletes. In some cases, it is not entirely clear whether a publication is based on a purposefully designed study protocol or whether scientists have taken the opportunity to observe ongoing activities or, perhaps, even a combination of the two.

Much focus has been given to high-altitude training and its effects on Hb_{mass} , for the simple reason that a high Hb_{mass} is an important factor for endurance-exercise performance, is known to be influenceable by hypoxia and is relatively easy to measure. Before delving into altitude training and Hb_{mass} data, it should also be stated clearly that Hb_{mass} is only one component of the highly complex nature of exercise performance, which is, in our opinion, very superficially understood. In some studies, Hb_{mass} might increase but might then not reveal the expected increase in $\dot{V}_{O_{2max}}$ or performance tests. Authors might then rightfully speculate that high-altitude training might facilitate haematological but not performance parameters. Our take on this, however, is that the continuous fine-tuning of carbon monoxide (CO) rebreathing methodology conducted over the last 20 years to determine Hb_{mass} allows for highly precise assessment of Hb_{mass} , if performed with the greatest of care, and also with the introduction of repeated measurements within a single measurement session (Lundby et al., 2023), whereas the variation for exercise performance tests is usually considerably larger, and thus requires greater statistical power/number of study participants to manifest into 'significant' differences. This might also highlight, however, that potential performance benefits of altitude training might not be substantial.

Recent studies on altitude training. A total of 31 studies published in the last decade are summarized in Table 1. Among the 19 controlled studies, 12 examined changes in Hb_{mass} , with seven (67%) reporting a positive effect of altitude training. Twelve studies evaluated changes in $\dot{V}_{O_{2max}}$, of which four (33%) reported improvements. Performance outcomes were assessed in 17 studies, with 8 (47%) demonstrating improvements following altitude exposure. It is worth noting that training load data are absent in 21% of the controlled studies (and 42% of the uncontrolled studies), limiting the ability to exclude confounding effects of training. Among the 12 uncontrolled studies, all 10 that assessed Hb_{mass} reported a positive response to altitude training. One of the four studies assessing $\dot{V}_{O_{2max}}$ found a positive effect, and three of the seven studies evaluating performance outcomes reported improvements (based on competition results in two of the three studies).

In summary, recent studies reveal substantial individual variability in responses to altitude training, with a majority indicating slight Hb_{mass} expansion, but fewer than half reporting performance benefits. Furthermore, the studies highlight the recurring challenge of monitoring and/or controlling training loads during training camps, which often hinders the drawing of robust conclusions about the real effects of altitude training.

Table 1. Studies on altitude training from 2015 to 2025

Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Controlled studies									
Garvican-Lewis et al. (2015)	<i>n</i> = 8	Distance runners Elite and non-elite	LHTH (<i>n</i> = 5) 1800 m 3 weeks	Same load between groups	YES (0 days post)	–	–	–	
Rodriguez et al. (2015)	<i>n</i> = 11	Swimmers Elite	LHTHL (<i>n</i> = 12)	23% higher load in the LHTHL group	NO (1–2 days post)	NO (1–2 days post)	–	YES (2 weeks post) YES (4 weeks post)	Performance based on 400 m swimming time trial
Brocherie et al. (2015)	<i>n</i> = 9	Hockey players Elite	LHTL (<i>n</i> = 12) 2500–3000 m 14 days (14 h/day)	Same load between groups	YES (2–3 days post) YES (3 weeks post)	–	–	YES (2–3 days post) YES (3 weeks post)	Performance based on Yo-yo test
Carr et al. (2015)	<i>n</i> = 10	Race walkers Elite	LHTL (<i>n</i> = 8) 3000 m 21 d (9 h/day)	No training data	YES	NO	NO	–	Data reported as relative changes (%) Performance based on time to exhaustion (no data in the control group) Control group derived from a previous study Post-altitude assessment conducted at an unspecified time

(Continued)

Table 1. (Continued)

Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Hauser et al. (2016)	<i>n</i> = 7	Triathletes Non-elite	LHTL (<i>n</i> = 11 + 10) 2250 m 18 days (13–17 h/day)	Same load between groups	YES (2–3 days post)	–	–	NO (2–3 days post)	Performance based on 3 km running time trial and cycling maximal aerobic power Altitude effect on performance deemed likely to be beneficial when using magnitude-based inference
Bejder et al. (2017)	<i>n</i> = 7 Cross-over study	Triathletes Non-elite	LHTL (<i>n</i> = 7) 2500–3500 m 6 weeks (8 h/day)	Same load between groups	–	–	–	NO (1 week post) NO (2 weeks post)	Double-blind, cross-over study Performance based on peak power during incremental test (1 week post) and 26 km cycling time trial (2 weeks post)
Inness et al. (2017)	<i>n</i> = 8	Australian football Non-elite	LHTL (<i>n</i> = 7) 3000 m 19 days (12 h/day)	Same load between groups	–	–	–	YES (1–3 days post)	Performance based on Yo-Yo test and on 1 and 2 km time trials Hb _{mass} not measured in control subjects Statistics derived from magnitude-based inference
Park et al. (2017)	<i>n</i> = 10	Distance runners Non-elite	LHTL (<i>n</i> = 10) 3000 m 4 weeks	No training data	–	–	YES (1 day post)	YES (1 day post)	Daily hypoxic exposure not reported Performance based on 3 and 5 km running time trials
Czuba et al. (2018)	<i>n</i> = 10	Cyclists Non-elite	LHTL (<i>n</i> = 10) 2100 m 3 weeks (11–12 h/day)	Same load between groups	–	YES (1 week post)	YES (1 week post)	YES (1 week post)	Performance based on 30 km cycling time trial

(Continued)

Table 1. (Continued)

Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Robach et al. (2018)	<i>n</i> = 8	Cross-country skiers Non-elite	LHTL (<i>n</i> = 11) 2200 m 26 days (17 h/day)	Same load between groups	NO (3 days post) NO (10 days post) NO (19 days post)	NO (3 days post) NO (10 days post) NO (19 days post)	NO (3 days post) NO (10 days post) NO (19 days post)	NO (3 days post) NO (10 days post) NO (19 days post)	Performance based on 3 km running time trial
Schmitt et al. (2018)	<i>n</i> = 6	Nordic skiers Elite	LHTL (<i>n</i> = 9) 2700 m 15 d (14 h/day)	Lower load in the LHTL group	–	NO (1 day post)	NO (1 day post)	NO (21 days post)	Performance based on 10 km roller-ski time trial, with data reported as relative changes (%)
Park et al. (2019)	<i>n</i> = 12	Runners Non-elite	LHTL (<i>n</i> = 12) 3000 m 3 weeks (12 h/day)	No training data	–	YES (1–7 days post)	YES (1–7 days post)	YES (1–7 days post)	Performance based on 3 km running time trial
Sitkowski et al. (2019)	<i>n</i> = 7	Cyclists Elite	LHTL (<i>n</i> = 8) 2200 m 3 weeks (>13 h/day)	Same load between groups	NO (13–15 days post)	NO (13–15 days post)	NO (13–15 days post)	NO (13–15 days post)	Double blind study, including placebo (<i>n</i> = 7) and control groups (<i>n</i> = 7) Performance based on maximal power during $\dot{V}O_{2\max}$ test Performance based on 5 km time trial
Meng et al. (2021)	<i>n</i> = 15	Rowers Non-elite	LHTH (<i>n</i> = 18) 2280 m 8 weeks	Same load between groups	–	YES (3 weeks post)	–	YES (3 weeks post)	
Kettunen et al. (2023)	<i>n</i> = 14	Cross-country skiers Elite	LHTLH (<i>n</i> = 15) 2250–2500 m 28 days (18 h/day)	Same load between groups	YES (3 days post)	NO (3 days post)	–	NO (3 days post)	Performance based on time to exhaustion
Peltonen et al. (2024)	<i>n</i> = 14	Various endurance sports Elite and non-elite	LHTH and LHTLH (<i>n</i> = 22) 1800–2585 m (various sites)	No training data	YES (1–2 days post) YES (30 days post)	YES (1–2 days post) NO (30 days post)	–	NO (1–2 days post) NO (30 days post)	Performance based on time to exhaustion

(Continued)

Table 1. (Continued)

Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Urianstad et al. (2024)	<i>n</i> = 12	Cyclists Non-elite	LHTL (<i>n</i> = 10) 2100 m 3 weeks	Same load between groups	NO (2 days post)	NO (2 days post)	NO (2 days post) NO (8 days post)	YES (2 days post) YES (8 days post)	Performance based on maximal power during $\dot{V}O_{2\max}$ test Heat suit training during the post-altitude phase in a subgroup of cyclists (Rønnestad et al., 2025)
Kuorelahti et al. (2025)	<i>n</i> = 8	Nordic skiers Elite and non-elite	LHTL (<i>n</i> = 7) 2500 m 21 days (18 h/day)	Same load between groups	YES (1–2 days post)	NO (1–2 days post)	–	NO (1–2 days post)	Performance based on time to exhaustion
Carin et al. (2025)	<i>n</i> = 9 Self-controlled study	Open water swimmers Elite	LHTH (<i>n</i> = 9) 1850 m 4 weeks	10% higher load in the LHTH group	YES (1 day post) YES (10 days post)	–	–	NO (10 days post)	Performance based on 400 m swimming time trial
Uncontrolled studies									
Sharma et al. (2018)	–	Runners Elite	LHTH (<i>n</i> = 8) 2100 m 3–4 weeks	Increase in load during the camp	YES (1–2 days post)	–	–	YES (1–8 days post)	Performance based on competition results
Heikura et al. (2018)	–	Runners and racewalkers Elite	LHTH (<i>n</i> = 38) 2135 m 3–4 weeks	Training record over 7 days	YES (0 days post)	–	–	–	Higher increase in Hb _{mass} in women compared with men
Yan et al. (2021)	–	Mixed martial art fighters Non-elite	LHTH (<i>n</i> = 9) 2418 m 3 weeks	No training data	YES (2 days post) (6 days post)	YES (4 days post) (9 days post)	–	YES (4 days post) (15 days post)	Performance based on 3 km running time trial

(Continued)

Table 1. (Continued)

Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Nummela et al. (2021)	–	Various sports <i>n</i> = 59 (82 stays) Elite	LHTH (<i>n</i> = 59) 1350–2500 m 16–42 days (various sites)	No training data	YES (1–3 days post)	–	–	–	Increase in Hb _{mass} in 56% of cases (all athletes), rising to 65% when the altitude camp was conducted above 2000 m
Skattebo and Hallen (2022)	–	Various sports <i>n</i> = 9 (34 stays) Elite	LHTH (<i>n</i> = 9) 1850–2320 m 18–23 days	No training data	YES (1–2 days post)	–	–	–	
Cerda-Kohler et al. (2022)	–	Rowers Elite	LHTH (<i>n</i> = 11) 2900 m 3 weeks	No training data	–	NO (14 days post)	YES (14 days post)	–	
Krumm et al. (2024)	–	Racewalkers Elite	LHTH (<i>n</i> = 10) 1720 m 4 weeks	Increase in load during the camp	YES (0 days post) NO (2 weeks post)	–	–	YES (3 weeks post)	Performance based on competition results
Mujika et al. (2024)	–	Middle-/long- distance swimmers Elite	LHTH (<i>n</i> = 7) 2320 m 3 weeks	Stable load during the camp	YES (0 days post)	–	–	NO 1 day post NO (2 weeks post)	Performance based on competition results
Yi et al. (2024)	–	Runners Non-elite	LHTHL (<i>n</i> = 16) 2800 m 3 weeks (10 h/day)	No training data	–	NO (1–2 days post)	NO (1–2 days post)	NO (1–2 days post)	Performance based on 5 km running time trial
Astridge et al. (2024)	–	Swimmers Elite	LHTH (<i>n</i> = 14) 2320 m 3 weeks	Increase in load during the camp	YES (7 days post) YES (14 days post)	–	–	NO (3 weeks post)	Performance based on competition results

(Continued)

Table 1. (Continued)									
Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Rønnestad et al. (2025)	–	Cyclists Elite and non-elite	LHTH (n = 18) 2100 m 3 weeks	Increase in load during the camp	YES (2 days post)	–	–	–	Carbon-monoxide inhalations during LHTH in a subgroup of cyclists (Urianstad et al., 2024)
Cubel et al. (2025)	–	Cyclists Elite	LHTH (n = 12) 3000 m 21 days	Increase in load during the camp	YES (0 days post) NO (10 days post)	NO (2 days post) NO (10 days post)	–	NO (2 days post) NO (10 days post)	Performance based on 20 min cycling time trial
This table summarizes the main studies on altitude training conducted over the past decade. The list might not be exhaustive. We identified relevant studies published from May 2015 to July 2025 on PubMed. We used combinations of the subject headings ‘hypox’, ‘altitude’, ‘live high’, ‘live low’, ‘train* high’, ‘train* low’ in healthy subjects, excluding studies on animal models, untrained participants and published in a language other than English. We included original articles that addressed LHTH, LHTL and LHTHL in trained individuals, reported pre- and postintervention measurements of Hb _{mass} , $\dot{V}O_{2\max}$ and/or endurance performance. Published material such as case reports, editorials and letters to the editor were not included. To avoid duplication, data originating from the same cohort of participants and published across multiple articles are presented only once. For controlled studies, effects are not reported when data are not available in the control group. Abbreviations: Hb _{mass} , haemoglobin mass; LHTH, live high–train high; LHTHL, live high–train high and low; LHTL, live high–train low; post, post-altitude measurements; $\dot{V}O_{2\max}$, maximal oxygen uptake. YES/NO indicates significant/no significant positive effects of altitude training, in comparison to sea-level training, demonstrated either by significant within-group improvements in the altitude group or, more rarely, by significant between-group differences.									

Table 2. Studies on heat exercise training, investigating its effect on haemoglobin mass

Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Controlled studies									
Oberholzer et al. (2019) and Mikkelsen et al. (2019)	$n = 9$	Amateur cyclists with moderate $\dot{V}O_{2\max}$	5½ weeks, 5 × 60 min sessions in heat/week	Same load between groups	NO	NO	NO	–	Tendency towards a higher increase in Hb _{mass} in HEAT compared with CON (time × group interaction: $P = 0.061$)
Rønnestad et al. (2021)	$n = 11$	Amateur cyclists with high $\dot{V}O_{2\max}$	5 weeks, 5 × 60 min sessions in heat/week	Same load between groups	YES	NO	NO	NO	Young cyclists with very high $\dot{V}O_{2\max}$. Small to intermediate effect sizes favouring heat for lactate threshold power output, gross economy in the fatigued state and 15 min mean power
Rønnestad et al. (2022b)	$n = 12$	Cross-country skiers with high $\dot{V}O_{2\max}$	5 weeks, 5 × 50 min weekly heat-suit training	Same load between groups	YES	NO	NO	NO	–
Rønnestad et al. (2022c)	$n = 10$	Amateur cyclists with high $\dot{V}O_{2\max}$	5 weeks, 5 × 50-min sessions/week in a heat chamber ($n = 12$) or while wearing a heat suit ($n = 10$)	Same load between groups	YES	NO	NO	NO	Heat chamber and heat suit equally effective. When groups were pooled ($n = 22$), this revealed a greater improvement in performance index (composed of power output at 4 mmol/L blood lactate, $W_{\max}$ and 15 min power output) than controls Hb _{mass} can be maintained high with lower stimuli, i.e. 3 weekly sessions

(Continued)

Table 2. (Continued)									
Reference	Control group	Sport Elite/non-elite	Intervention	Training load	Hb _{mass}	$\dot{V}O_{2\max}$	Submaximal indices	Performance	Observations
Lundby et al. (2023)	<i>n</i> = 23	Amateur cyclists with high $\dot{V}O_{2\max}$	5 weeks, 5 × 50 min sessions/week	Same load between groups	YES	YES	YES	YES	Female response similar to male Forty-seven subjects in total, all with very high $\dot{V}O_{2\max}$
Cubel et al. (2024)	<i>n</i> = 10	Continental Team cyclists	5 weeks, 6 × 60 min sessions in heat/week	Same load between groups	YES	NO	NO	–	Hb _{mass} was increased by 30 g after 3 weeks and by 40 g after 5 weeks Heat improved incremental peak power output without changes in O ₂ uptake, which was a non-significant numerical 180 ml/min higher from pre- to postintervention
This table summarizes the main studies on heat exercise training. The list might not be exhaustive. We identified relevant studies published until July 2025 on PubMed. We used combinations of the subject headings ‘heat’, ‘training’ and ‘exercise’. Published material such as case reports (Rønnestad et al., 2022a), editorials and letters to the editor were not included. Abbreviations: Hb _{mass} , haemoglobin mass; $\dot{V}O_{2\max}$, maximal oxygen uptake. <i>W</i> _{max} , Maximal workrate; YES/NO indicates significant/no significant positive effects of heat exercise training, in comparison to the control group demonstrated by significant between-group differences.									

Variation in response to high-altitude training: responders vs. non-responders and its reproducibility.

In the (by now classic) high-altitude training study by Levine and Stray-Gundersen (1997), they observed that the increase in estimated red blood cell volume (derived from plasma volume measurements) varied among study participants and, accordingly, termed these 'responders' or 'non-responders'. Such variation might perhaps be somewhat surprising, at least at the upstream cellular level, because nearly all human cells possess the hypoxia-inducible factor-1 system that is responsible for activating erythropoiesis. The variation might obviously stem downstream from hypoxia-inducible factor-1 activation, but might be one example of cellular and phenotypical responses not necessarily going hand in hand.

Accordingly, Levine's group has reported considerable variation in the erythropoietin (EPO) response to acute hypoxic exposure (Ge et al., 2002). Regardless, the variation in Hb_{mass} response to high-altitude training has been the target for investigation by two research groups, in particular, who have assessed data gathered during national team camps (Nummela et al., 2021; Skattebo & Hallen, 2022). In these studies, however, sea-level control training is lacking, and it cannot be ruled out that the variations observed would have been similar if studied at sea level. The Hb_{mass} is not a steady value, with an average daily turnover of erythrocytes of ~1%, and this number can obviously vary depending on many factors.

Nummela et al. (2021) set out to determine whether athletes can be classified as responders or non-responders. The total dataset included data from an impressive 82 high-altitude training camps and with the inclusion of 59 athletes. The degree of altitude ranged from 1350 to 2500 m. The data revealed that Hb_{mass} was higher in 56% of all participants in comparison to pre-altitude training (but given that there was no control group, not in comparison to sea-level control training) and that this number was increased to 65% when only camps conducted above 2000 m a.s.l. were considered. In a subgroup of 15 athletes participating at least twice in camps, they found that 27% always had a positive Hb_{mass} response following a high-altitude training camp, whereas 13% always a negative response and 60% had both a positive and a negative response. Accordingly, the authors concluded that this 'does not support classification of an athlete permanently as a responder or non-responder', but nonetheless support the responder vs. non-responder concept as such, but which, in a strict sense, could also be the result of external/ biological factors, such as fluctuations over time. Intriguingly, Robertson et al. (2010) demonstrated some 15 years ago that although group mean response could be replicated for Hb_{mass} between two identical high-altitude training camps, this was not the case at the individual level. Of note, exercise

performance in that study could not be reproduced even at the group mean level.

Realizing that Hb_{mass} fluctuates across a sport season, Skattebo and Hallen (2022) tested whether a low individual baseline measure of Hb_{mass} would lead to a greater response in Hb_{mass} when national team athletes participated in a high-altitude training camp (i.e. observational study with no control group). They did so by calculating the individual mean baseline for Hb_{mass} of three to five high-altitude training camps, then investigated whether the percentage deviation from that individual mean baseline affected the altitude-induced Hb_{mass} response. The conclusion from the study is that an initially low Hb_{mass} before sojourn affects the high-altitude response in a positive manner. Whether this is caused by hypoxia as such or whether it exemplifies the natural variation reported, being 3–5% over 10–12 months (Garvican et al., 2010; Prommer et al., 2008), remains uncertain. Of note, Peltonen et al. (2024) did not find an association between initial sea-level baseline values and the response at high altitude. However, given that they also included the sea-level group Hb_{mass} data in their calculation, the validity of their observation is questionable. Nevertheless, although we and others have previously suggested an association between the initial level of Hb_{mass} and its potential expansion with high-altitude training (Koivisto-Mørk et al., 2021; McLean et al., 2013; Robach & Lundby, 2012; Skattebo & Hallen, 2022), the existence of a 'physiological ceiling' for Hb_{mass} remains difficult to explain and is not always confirmed. Observational data (without a sea-level control group) indicate that even elite cyclists in the competition period can slightly increase their Hb_{mass} following altitude training (Cubel et al., 2025).

In the same study population as used by Skattebo and Hallen (2022), Koivisto-Mørk et al. (2021) analysed whether baseline values for serum ferritin in athletes with clinically normal iron stores and iron supplementation during the camp had an effect on the Hb_{mass} response. No associations were found, however. In a similar manner, Nummela et al. (2021) found that the Hb_{mass} response to high-altitude training above 2000 m increased from 65 to 69% when only those with normal ferritin levels were included, suggesting a small negative effect if ferritin levels are lower than normal.

Although responder vs. non-responder seems mostly to be discussed in relationship to athletes participating in altitude training, one might ask whether this phenomenon is common also among healthy humans exposed to high altitude? When exposing eight healthy non-sport-active individuals to 3450 m altitude for 1 month, we observed Hb_{mass} to increase on average by 5.3%, but also that the magnitude of increase ranged from 2.5 to 11.1%, i.e. from what is barely detectable with the CO rebreathing method to a 'sizeable' increase (Siebenmann et al., 2015).

In a recent study, we had the opportunity to determine Hb_{mass} in response to 1 year of exposure to 3800 m high altitude on the Antarctic continent, with the inclusion of an according sea-level control group. Measurements were performed before departure, then again every month during the 1 year exposure. Following 1 year of exposure, the average increase in Hb_{mass} in the high-altitude group was 18%, but which ranged from 1 to 49% (Robach et al., 2025). To the best of our knowledge, this is the largest increase in Hb_{mass} observed at the individual level and clearly documents the great potential that prolonged altitude exposure might have for athletic purposes. Had we not conducted monthly measurements, allowing us closely to observe precise response kinetics throughout the full intervention year, such variation and degree of response would have been difficult to understand. However, owing to this monthly sampling over that entire year, the study clearly demonstrates the existence of non-responders, at least with 1 year exposure to 3800 m. Unfortunately, our study does not provide information on the individual reproducibility of this finding. Despite considerable efforts (iron stores, EPO response and genetic analysis), the source of variation remains difficult to identify in the study, however. Could we have foreseen these dramatic variations in response?

The high-altitude responder/non-responder phenomenon might perhaps not be all that different from human adaptations to other external stimuli. One such example could be exercise training, where a given training dose can cause marked different increases in exercise performance. In one such study, $\dot{V}_{O_{2max}}$ is reported to increase by 3–60%, although training was identical among study participants (Montero & Lundby, 2017). In that study, participants who trained less frequently per week, possibly reflecting an insufficient ‘training dose’ comparable to ‘too’ low an altitude dose, included some non-responders to exercise training, whereas no non-responders were found among those training more than four times per week, a pattern that might be comparable to the effects of higher altitude exposure or longer exposure duration. It is perhaps easy to imagine that non-responders to a given degree of hypoxia might respond if the dose and/or degree is increased, although this remains unexplored.

The obvious consequence of this would be to identify altitude responders (to altitudes where athletes can still live and train), then prolong the stimulus for as long as deemed possible, but at the same time without negatively affecting other parts of athletes’ lives.

Attempts to prolong the hypoxic effect on Hb_{mass} . A consistent observation is that Hb_{mass} decreases back to baseline values within 1–2 weeks following high-altitude

exposure (Klein et al., 2021; Siebenmann et al., 2015), which is in stark contrast to the expected lifespan of 120 days of a red blood cell. Based on animal experiments, an initial explanation given for this phenomenon was a selective destruction of the newest synthesized blood cells, accordingly termed ‘neocytolysis’ (Song et al., 2017). More recently, in humans it was demonstrated that erythropoiesis was essentially blocked upon return to sea level (Klein et al., 2021), which would decrease Hb_{mass} by $\sim 1\%$ /day if terminated completely (theoretically meaning ~ 50 g Hb_{mass} in a week in a person with an Hb_{mass} of 750 g). It could thus be speculated that an intervention/stimulus facilitating circulating EPO upon return to sea level could limit or entirely abolish the loss of Hb_{mass} upon return to sea level. Two strategies have been exploited thus far, with positive outcomes. First, Peltonen’s research team found that in 22 national team athletes (from mixed disciplines) performing (unsupervised) intermittent hypoxic exposure (2 h rest in normobaric hypoxia) and hypoxic exercise (1 h at low intensity) every third day for 1 month following 1 month of high-altitude training (various protocols), the elevated Hb_{mass} was maintained, whereas this was not the case for the 22 athletes in the control group who simply returned to sea level, with no attempts made to preserve the elevated Hb_{mass} . Second, in a similar manner, Rønnestad et al. (2025) showed that three weekly heat-training sessions (see below) performed over 3.5 weeks is an effective strategy for maintaining the elevated Hb_{mass} in a mix of under-23 national team mountain bikers, enthusiastic cyclists and sport students following a high-altitude training camp, compared with control subjects simply ‘returning home’. From a strict scientific point of view, both studies suffer somewhat from not being strictly designated research projects with random group assignments, but rather opportunistically following ongoing behaviour of athletes, with lack of training data in one study (Peltonen et al., 2024). Nonetheless, what can be learned from these studies is that it might be possible to maintain an elevated Hb_{mass} following high-altitude training. Whether this is related to an intervention-induced spike in circulating EPO remains unknown, because, in the study by Rønnestad et al. (2025), no attempts were made to delve into the physiological mechanisms, and, in contrast, no EPO spike was observed in the study by Peltonen et al. (2024), which could have been related to timing of the blood sampling. Other strategies, such as postural manipulations (Montero et al., 2016), might also provide at least some degree of erythropoietic stimulus, and augmented ketone levels may or may not (Evans et al., 2023; Kanta et al., 2025; Poffé et al., 2023; Stalmans et al., 2025), and these will probably be tested after high-altitude training camps in the future.

Providing a hypoxic response by means of carbon monoxide inhalation. Since our last review (Lundby & Robach, 2016), publications have emerged that have used the strategy of CO administration to mimic high-altitude exposure. The underlying rationale is that CO inhalation facilitates the synthesis of EPO (Montero & Lundby, 2019) and that smokers have higher [Hb] than non-smokers (Nordenberg et al., 1990). In order to provide a more or less constant degree of hypoxia, CO has to be inhaled several times per day, because its half-life is ~ 5 h (Lombardi et al., 2013; Siebenmann et al., 2017). The strategy was first tested by a team of US and German scientists (Ryan et al., 2016), showing that one daily inhalation for 10 days did not increase Hb_{mass} , which might come as no big surprise because this phenotype has been shown to be a lengthy process. Later, the same team (Schmidt et al., 2020) administered CO five times per day for 3 weeks and observed an increase in Hb_{mass} and in exercise performance. Some years later, Urianstad et al. (2024) tested whether the application of CO to a mix of Norwegian national team members and amateur cyclists/university sport students (same study as the paper by Rønnestad et al., 2025) would increase Hb_{mass} more than if not inhaling CO during a high-altitude training camp. Given that the hypoxic stimulus became enhanced, the expected outcome was demonstrated, but again eliciting rather large individual variations in outcome (3–8% increase). The study by Urianstad et al. (2024) led to a series of events, finally leading the International Cycling Union (UCI) to ban the use of CO for purposes other than measurement of Hb_{mass} (i.e. diagnostic purposes). The latter is fully in line with what we have proposed previously (Lundby & Robach, 2024), hence we are happy with this decision.

Heat exercise training

Acclimatization to a warm environment, whether combined with exercise training or not, has been known for decades to increase sweat rates and expand plasma volume, which are also associated with an increased exercise performance in such environments. In addition, daily 45–90 min exercise-training sessions performed for 10–14 days in a warm environment, but while living in a thermoneutral environment (i.e. heat training), has also been shown to expand plasma volume (Périard et al., 2021). It was while performing such a 14-day-long heat-training study (Keiser et al., 2015) that we came to speculate whether the heat-training-induced increase in plasma volume, and the consequent lowering of the haematocrit, could stimulate the production of red blood cells through the proposed ‘critmeter’ function (Dunn et al., 2007) of the kidney (Fig. 1). We reasoned that the previously performed heat-training studies lasting

10–14 days would be too short to increase red blood cell volume/ Hb_{mass} , because in previous studies where we had injected supraphysiological concentrations of recombinant EPO (Robach et al., 2021) or used altitude exposure (Siebenmann et al., 2015), a Hb_{mass} response had first been observed after 2 and 3 weeks, respectively. In the first performed heat-training study with the intent to expand Hb_{mass} (Oberholzer et al., 2019), we therefore opted for a 5-week intervention period and used moderately fit individuals, rather than elite athletes, with the rationale still to allow for adaptations. With five weekly training sessions, each lasting 60 min and performed at 60% of $\dot{V}_{\text{O}_{2\text{max}}}$, we observed an increase in Hb_{mass} that did not alter exercise performance, probably related to an underpowered study. As mentioned above, the variation associated with Hb_{mass} assessment is typically lower than the variation for exercise performance testing. In terms of physiological mechanisms for the Hb_{mass} expansion, not much was learned from this first study, because most analysis of blood volume-regulating hormones remained unchanged, which might have been related to missing a potential alteration owing to a narrow/limited sampling window.

As a next step, we tested whether a similar protocol, i.e. five weekly heat-training sessions, each lasting 45 min and performed at a rate of perceived exertion of $\sim 13/20$ (‘somewhat hard’ on the Borg scale), for 5 weeks, but performed in actively trained cyclists, would have similar effects on Hb_{mass} . Although Hb_{mass} was increased by 46 g, the $\dot{V}_{\text{O}_{2\text{max}}}$, lactate threshold and 15 min time-trial all-out exercise performance remained statistically unchanged (Rønnestad et al., 2021). There were, however, small to intermediate effect sizes favouring heat exercise training for lactate threshold, gross economy (in a fatigued state) and 15 min mean power.

The next step for us was to test whether such a protocol could be used to increase Hb_{mass} and exercise performance in highly fit individuals, hence competing Norwegian cross-country skiers where enrolled into yet another study, which revealed a 30 g greater Hb_{mass} , but, in contrast, no group differences were observed for changes in work economy, running velocity, $\dot{V}_{\text{O}_{2\text{max}}}$ or 15-min running distance (Rønnestad et al., 2022b). Thus, in initial heat exercise training studies, we could typically detect an increase in Hb_{mass} , but with only very limited effects on exercise performance parameters. However, by increasing subject numbers, we were able to induce an increase Hb_{mass} and in exercise performance, in trained individuals (Lundby et al., 2023; Rønnestad et al., 2022c).

Studies performed on this topic by our group(s) (Table 2) have also revealed that: (i) following the standard 5 weeks with five weekly heat-training sessions, Hb_{mass} can continue to increase, or at least remains stable, by performing three heat-training sessions per week (Rønnestad et al., 2022c); (ii) exercise training

conducted either in a warm environment or simply by overdressing elicits similar improvements in Hb_{mass} ; (iii) the responses observed are similar between female and male subjects (Lundby et al., 2023, Rønnestad et al., 2022c) (today, the overdressing approach seems to be favoured by athletes for practical reasons, in comparison to performing the training in a warm environment); in the later study (Lundby et al., 2023), we obtained skeletal muscle biopsies to investigate whether heat training would induce mitochondrial biogenesis and angiogenesis, potentially through heat shock protein activation, but neither response occurred; and (iv) the response in Hb_{mass} seems to continue to increase with 8 weeks of heat training

in national team athletes (Rønnestad et al., 2022a). More recently, with Lars Nybo's group, we have demonstrated that 3 weeks with heat training is sufficient to increase Hb_{mass} in cyclists (Continental Team level) (Cubel et al., 2024), but also that the magnitude is numerically higher after 5 weeks.

Thus, since 2019 we have consistently demonstrated an increase in Hb_{mass} with heat training, which also seems to drive exercise performance in studies with sufficient statistical power, and it would seem that heat training could be a promising substitute for the more classic altitude training (Nybo et al., 2024). Indeed, we have tested the efficacy of live high–train high altitude training

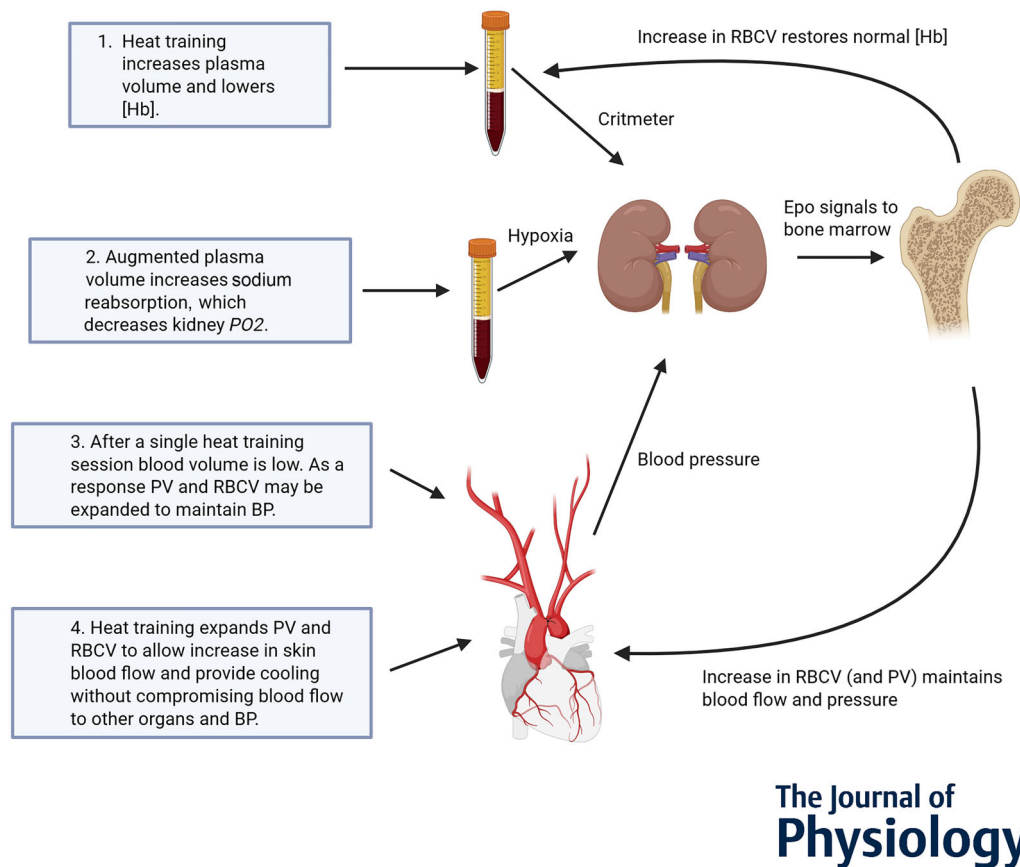


Figure 1. Proposed mechanisms for red blood cell volume/total haemoglobin mass to become augmented with heat training

Pathway 1 illustrates the hypothesis that when repeated heat-training sessions have resulted in an augmented plasma volume, that this might be registered by the kidney through its proposed 'critmeter' function and thereby red blood cell volume might be increased in an attempt to maintain [Hb] within a normal range (Dunn et al., 2007). Pathway 2 suggests that red blood cell volume might become elevated as a consequence of kidney tissue hypoxia. Repeated heat training increases plasma volume, which might facilitate kidney sodium reabsorption, hence kidney work, which decreases kidney tissue PO_2 . Pathway 3 suggests that after each single session of heat training, plasma volume might acutely be reduced. This could lead to postexercise hypotension, and red blood cell volume (and plasma volume) could become elevated to maintain blood pressure. Pathway 4 suggests that red blood cell volume (and plasma volume) might be augmented simultaneously, to allow blood flow to be directed to the skin for cooling during exercise, while, at the same time, maintaining perfusion of other tissues without compromising blood pressure. None of the four suggested pathways to increase red blood cell volume with heat training has yet been verified. Abbreviations: BP, blood pressure; Epo, erythropoietin; [Hb], haemoglobin concentration; PO_2 , oxygen partial pressure; PV, plasma volume; RBCV, total red blood cell volume.

(3000 m a.s.l. for 21 days) compared with 5 weeks of heat training (6×1 h sessions/week) in the same seven cyclists (Continental Team level) and found that the increase in Hb_{mass} was similar between these two strategies (Cubel et al., 2025). Although, from an exercise performance point of view, the data from this study seem to favour heat training, it is important to emphasize that the post-altitude training performance tests were performed in a state with subjects being jetlagged, which might have muddled the performance results.

Of note, to date no long-term heat-training-specific study seems to have been conducted outside Scandinavia or with the involvement of other research groups. These are obviously great study limitations, and it might therefore be questioned whether heat training might also be effective in regions of the world where ambient temperatures are higher than in Scandinavia, and if performed by other research teams.

Despite initial attempts to unravel the underlying mechanisms facilitating the expansion of Hb_{mass} with heat training, thus far these have been in vain, and we can only speculate that the expansion might be related to our initial hypothesis (Keiser et al., 2015), i.e. secondary to an expanded plasma volume and according drop in haematocrit, as also outlined elsewhere (Montero & Lundby, 2018). However, this might be entirely naïve, and other mechanisms might well be involved (Fig. 1), including: (pathway 1) a greater blood volume, allowing for a greater blood perfusion of skin during heat exposure, in turn providing a greater cooling effect, without compromising blood pressure; (pathway 2) a decrease in kidney blood flow during exercise in a warm environment, as blood flow is directed towards skin to provide cooling, whereby the decrease in kidney blood flow could result in local kidney hypoxia and thus provide a signal for augmented EPO synthesis; (pathway 3) a heat-training-induced reduction in plasma volume, decreasing central venous pressure, which has been shown to facilitate EPO synthesis (Montero et al., 2016); and/or (pathway 4) heat-training-induced plasma volume contraction, causing upregulation of sodium reabsorption and augmented O_2 use, which results in local hypoxia in the kidney and thereby EPO synthesis.

Heat training has quickly become highly popular amongst athletes, as seen on YouTube, with producers of gadgets that might be more or less useful for monitoring different parameters associated with heat training. Based on these and looking back, the word 'training' in the context of heat training seems to cause confusion, because some advocates of heat training have proposed that the training during heat exposure should be hard and even of the interval type of intensities. Our take on this is that the 'training' or, perhaps better, 'physical activity' is a simple and highly effective way of increasing core temperature to a greater extent than if simply being exposed passively

to heat (i.e. without exercise), but also that not much exercise intensity is needed in a warm environment or in an environment with poor cooling ability, in order to drive core temperature, hence sweat rate, up. In our practical experience (including Pro and World Tour cyclists and Olympic calibre runners and skiers), heat training should be conducted at low enough exercise intensities to increase core temperature, but not at intensities that are normally considered as training (for athletes), with the intent not to interfere with the normal training regimes of athletes. The intent, in our opinion, is to increase core temperature (by whatever means) to facilitate an elevated sweat rate. The acute loss of sweat is, we speculate, what acutely lowers plasma volume, which ultimately might drive the chronic expansion of plasma volume and increase in Hb_{mass} . Thus, an easy manner to determine whether a heat-training session has been successfully performed or not is thus simply by weighing the athlete before and after each heat-training session and note the (mainly) fluid loss. We recommend aiming at an initial 1 kg loss (a little less for small individuals, a little more for bigger individuals) for each session and adjustment according to athlete feedback from there. For this, no app or special sensor/watch is needed.

Altitude exposure combined with heat training?

From a theoretical standpoint, the fact that exposure to either altitude or heat generally leads to an increase in Hb_{mass} suggests that combining the two stressors might have a synergistic effect on Hb_{mass} . This concept is currently gaining interest in elite sports. However, given that the two stressors have opposing effects on plasma volume (an increase with heat exposure *vs.* a decrease at altitude), the existence of a synergistic effect has yet to be established, as suggested by a recent review (Gibson et al., 2017). To date, there is no compelling evidence demonstrating an advantage from combining heat and hypoxia to boost Hb_{mass} and/or performance, although we have practical knowledge from ProTour cyclists that such an approach is feasible and that the outcome is perceived as 'rather positive'. This question deserves further investigation.

Conclusion

High-altitude training remains a popular training strategy to implement in the preparation of elite endurance athletes for major events. Both the haematological and performance benefits from such an approach are highly variable, and its reproducibility is questionable. In contrast, it seems that the haematological adaptations associated with heat training are far more robust, but also that the performance associated gains are small. Thus,

heat training should perhaps not be implemented before all other steps involved in elite training are in place and, importantly, heat training should not be performed in such a manner that it might limit other aspects of the life of elite athletes.

References

- Astridge, D. J., McKenna, M., Campbell, A., & Turner, A. P. (2024). Haemoglobin mass responses and performance outcomes among high-performance swimmers following a 3-week live-high, train-high camp at 2320 m. *European Journal of Applied Physiology*, **124**(8), 2389–2399.
- Bejder, J., Andersen, A. B., Buchardt, R., Larsson, T. H., Olsen, N. V., & Nordsborg, N. B. (2017). Endurance, aerobic high-intensity, and repeated sprint cycling performance is unaffected by normobaric “Live high-train Low”: A double-blind placebo-controlled cross-over study. *European Journal of Applied Physiology*, **117**(5), 979–988.
- Bejder, J., & Nordsborg, N. B. (2018). Specificity of “live high-train low” altitude training on exercise performance. *Exercise and Sport Sciences Reviews*, **46**(2), 129–136.
- Brocherie, F., Millet, G. P., Hauser, A., Steiner, T., Rysman, J., Wehrlin, J. P., & Girard, O. (2015). “Live high-train low and high” hypoxic training improves team-sport performance. *Medicine & Science in Sports & Exercise*, **47**(10), 2140–2149.
- Carin, R., Faucher, C., Connes, P., Viard, V., Martin, M., Ducray, M., Pla, R., Merino, M., Gaston, A. F., Lacroix, J., Doucende, G., Joly, P., Stauffer, E., Brugniaux, J., Vergès, S., Brocherie, F., Pichon, A., Robach, P., & Nader, E. (2025). Persisting elevation of total hemoglobin mass after altitude training in elite swimmers: a potential role of prolonged erythrocyte survival. *American Journal of Physiology-Heart and Circulatory Physiology*, **329**(4), H789–H800.
- Carr, A. J., Saunders, P. U., Vallance, B. S., Garvican-Lewis, L. A., & Gore, C. J. (2015). Increased hypoxic dose after training at low altitude with 9h per night at 3000m normobaric hypoxia. *Journal of Sports Science and Medicine*, **14**, 776–782.
- Cerda-Kohler, H., Haichelis, D., Reuquen, P., Miarka, B., Homer, M., Zapata-Gomez, D., & Aedo-Munoz, E. (2022). Training at moderate altitude improves submaximal but not maximal performance-related parameters in elite rowers. *Frontiers in Physiology*, **13**, 931325.
- Cubel, C., Fischer, M., Stampe, D., Klaris, M. B., Bruun, T. R., Lundby, C., Nordsborg, N. B., & Nybo, L. (2024). Time-course for onset and decay of physiological adaptations in endurance trained athletes undertaking prolonged heat acclimation training. *Temperature (Austin)*, **11**(4), 350–362.
- Cubel, C., Klaris, M. B., Larsen, J. V., Faiss, R., Nybo, L., & Lundby, C. (2025). Haematological adaptations to high-altitude and heat acclimation training in elite male cyclists. *Experimental Physiology*. Advance online publication. <https://doi.org/10.1113/EP092968>
- Czuba, M., Fidos-Czuba, O., Ploszczycza, K., Zajac, A., & Langfort, J. (2018). Comparison of the effect of intermittent hypoxic training vs. the live high, train low strategy on aerobic capacity and sports performance in cyclists in normoxia. *Biology of Sport*, **35**(1), 39–48.
- Dunn, A., Lo, V., & Donnelly, S. (2007). The role of the kidney in blood volume regulation: The kidney as a regulator of the hematocrit. *American Journal of the Medical Sciences*, **334**(1), 65–71.
- Evans, E., Walhin, J. P., Hengist, A., Betts, J. A., Dearlove, D. J., & Gonzalez, J. T. (2023). Ketone monoester ingestion increases postexercise serum erythropoietin concentrations in healthy men. *American Journal of Physiology-Endocrinology and Metabolism*, **324**(1), E56–E61.
- Faiss, R., Raberin, A., Brocherie, F., & Millet, G. P. (2024). Repeated-sprint training in hypoxia: A review with 10 years of perspective. *Journal of Sports Sciences*, **43**(19), 2164–2178.
- Garvican-Lewis, L. A., Halliday, I., Abbiss, C. R., Saunders, P. U., & Gore, C. J. (2015). Altitude exposure at 1800 m increases haemoglobin mass in distance runners. *Journal of Sports Science and Medicine*, **14**(2), 413–417.
- Garvican, L., Martin, D., McDonald, W., & Gore, C. (2010). Seasonal variation of haemoglobin mass in internationally competitive female road cyclists. *European Journal of Applied Physiology*, **109**(2), 221–231.
- Garvican, L. A., Saunders, P. U., Pyne, D. B., Martin, D. T., Robertson, E. Y., & Gore, C. J. (2012). Hemoglobin mass response to simulated hypoxia “blinded” by noisy measurement? *Journal of Applied Physiology* (1985), **112**(10), 1797–1798.
- Ge, R. L., Witkowski, S., Zhang, Y., Alfrey, C., Sivieri, M., Karlsen, T., Resaland, G. K., Harber, M., Stray-Gundersen, J., & Levine, B. D. (2002). Determinants of erythropoietin release in response to short-term hypobaric hypoxia. *Journal of Applied Physiology* (1985), **92**(6), 2361–2367.
- Gibson, O. R., Taylor, L., Watt, P. W., & Maxwell, N. S. (2017). Cross-adaptation: Heat and cold adaptation to improve physiological and cellular responses to hypoxia. *Sports Medicine*, **47**(9), 1751–1768.
- Hauser, A., Schmitt, L., Troesch, S., Saugy, J. J., Cejuela-Anta, R., Faiss, R., Robinson, N., Wehrlin, J. P., & Millet, G. P. (2016). Similar haemoglobin mass response in hypobaric and normobaric hypoxia in athletes. *Medicine & Science in Sports & Exercise*, **48**(4), 734–741.
- Heikura, I. A., Burke, L. M., Bergland, D., Uusitalo, A. L. T., Mero, A. A., & Stellingwerff, T. (2018). Impact of energy availability, health, and sex on haemoglobin-mass responses following live-high-train-high altitude training in elite female and male distance athletes. *International Journal of Sports Physiology and Performance*, **13**(8), 1090–1096.
- Inness, M. W. H., Billaut, F., & Aughey, R. J. (2017). Live-high train-low improves repeated time-trial and Yo-Yo IR2 performance in sub-elite team-sport athletes. *Journal of Science and Medicine in Sport*, **20**(2), 190–195.
- Kanta, J. M., Lundsgaard, A., Schaufuss, A., Kleinert, M., Kiens, B., & Fritzen, A. M. (2025). Induction of erythropoietin by dietary medium-chain triacylglycerol in humans. *American Journal of Physiology-Endocrinology and Metabolism*, **328**(2), E210–E216.

- Keiser, S., Flück, D., Hüppin, F., Stravs, A., Hilty, M. P., & Lundby, C. (2015). Heat training increases exercise capacity in hot but not in temperate conditions: A mechanistic counter-balanced cross-over study. *American Journal of Physiology-Heart and Circulatory Physiology*, **309**(5), H750–H761.
- Keiser, S., Meinild-Lundby, A. K., Steiner, T., Trosch, S., Rauber, S., Krafft, A., Burkhardt, T., Hilty, M. P., Siebenmann, C., Wehrin, J. P., & Lundby, C. (2017). Detection of blood volumes and haemoglobin mass by means of CO re-breathing and indocyanine green and sodium fluorescein injections. *Scandinavian Journal of Clinical and Laboratory Investigation*, **77**(3), 164–174.
- Kettunen, O., Leppavuori, A., Mikkonen, R., Peltonen, J. E., Nummela, A., Wikstrom, B., & Linnamo, V. (2023). Hemoglobin mass and performance responses during 4 weeks of normobaric “live high-train low and high”. *Scandinavian Journal of Medicine & Science in Sports*, **33**(8), 1335–1344.
- Klein, M., Kaestner, L., Bogdanova, A. Y., Minetti, G., Rudloff, S., Lundby, C., Makhro, A., Seiler, E., van Cromvoirt, A., Fenk, S., Simionato, G., Hertz, L., Recktenwald, S., Schäfer, L., Haider, T., Fried, S., Borsch, C., Marti, H. H., Sander, A., & Mairbäurl, H. (2021). Absence of neocytolysis in humans returning from a 3-week high-altitude sojourn. *Acta Physiologica (Oxford, England)*, **232**(3), e13647.
- Koivisto-Mørk, A. E., Svendsen, I. S., Skattebo, Ø., Hallén, J., & Paulsen, G. (2021). Impact of baseline serum ferritin and supplemental iron on altitude-induced haemoglobin mass response in elite athletes. *Scandinavian Journal of Medicine & Science in Sports*, **31**(9), 1764–1773.
- Krumm, B., Vallance, B., Burke, L., Garcia, J., Bouten, J., Brocherie, F., Saugy, J. J., Botre, F., & Faiss, R. (2024). High-level performances following low altitude training and tapering in warm environments in elite racewalkers. *European Journal of Sport Science*, **24**(8), 1120–1129.
- Kuorelahti, T., Ihalainen, J. K., Linnamo, V., Badenhorst, C., Kettunen, O., & Mikkonen, R. S. (2025). Influence of “live high-train low” on haemoglobin mass and post-exercise hepcidin response in female endurance athletes. *European Journal of Applied Physiology*, **125**(9), 2503–2514.
- Levine, B. D., & Stray-Gundersen, J. (1997). “Living high-training low”: Effect of moderate-altitude acclimatization with low-altitude training on performance. *Journal of Applied Physiology*, **83**(1), 102–112.
- Lombardi, G., Lanteri, P., Fiorella, P. L., Simonetto, L., Impellizzeri, F. M., Bonifazi, M., Banfi, G., & Locatelli, M. (2013). Comparison of the haematological profile of elite road cyclists during the 2010 and 2012 GiroBio ten-day stage races and relationships with final ranking. *PLoS ONE*, **8**(4), e63092.
- Lundby, C., Hamarsland, H., Hansen, J., Bjørndal, H., Berge, S. N., Hammarstöm, D., & Rønnestad, B. R. (2023). Hematological, skeletal muscle fiber, and exercise performance adaptations to heat training in elite female and male cyclists. *Journal of Applied Physiology* (1985), **135**(1), 217–226.
- Lundby, C., & Robach, P. (2016). Does ‘altitude training’ increase exercise performance in elite athletes? *Experimental Physiology*, **101**(7), 783–788.
- Lundby, C., & Robach, P. (2024). Toxic trend: Inhalation of the silent killer carbon monoxide by athletes to increase haemoglobin mass. *Journal of Applied Physiology* (1985), **137**(4), 820.
- Lundby, C., Siebenmann, C., & Robach, P. (2012). Reply to Garvican, Saunders, Pyne, Martin, Robertson, and Gore. *Journal of Applied Physiology*, **112**(10), 1799.
- McLean, B. D., Buttifant, D., Gore, C. J., White, K., & Kemp, J. (2013). Year-to-year variability in haemoglobin mass response to two altitude training camps. *British Journal of Sports Medicine*, **47**(Suppl 1), i51–i58.
- Meng, Z., Gao, H., Li, T., Ge, P., Xu, Y., & Gao, B. (2021). Effects of eight weeks altitude training on the aerobic capacity and microcirculation function in trained rowers. *High Altitude Medicine & Biology*, **22**(1), 24–31.
- Mikkelsen, C. J., Junge, N., Piil, J. F., Morris, N. B., Oberholzer, L., Siebenmann, C., Lundby, C., & Nybo, L. (2019). Prolonged heat acclimation and aerobic performance in endurance trained athletes. *Frontiers in Physiology*, **10**, 1372.
- Montero, D., & Lundby, C. (2017). Refuting the myth of non-response to exercise training: ‘non-responders’ do respond to higher dose of training. *The Journal of Physiology*, **595**(11), 3377–3387.
- Montero, D., & Lundby, C. (2018). Regulation of red blood cell volume with exercise training. *Comprehensive Physiology*, **9**(1), 149–164.
- Montero, D., & Lundby, C. (2019). Arterial O₂ content regulates plasma erythropoietin concentration independently of arterial O₂ tension: A randomized controlled experimental study. *Kidney International*, **95**(1), 173–177.
- Montero, D., Rauber, S., Goetze, J. P., & Lundby, C. (2016). Reduction in central venous pressure enhances erythropoietin synthesis: Role of volume-regulating hormones. *Acta Physiologica (Oxford, England)*, **218**(2), 89–97.
- Mujika, I., Bourdillon, N., Zelenkova, I., Vergnoux, F., & Millet, G. P. (2024). Hematological and performance adaptations to altitude training (2,320 m) in elite middle-distance and distance swimmers. *Frontiers in Physiology*, **15**, 1474479.
- Nordenberg, D., Yip, R., & Binkin, N. J. (1990). The effect of cigarette smoking on haemoglobin levels and anemia screening. *Journal of the American Medical Association*, **264**(12), 1556.
- Nummela, A., Eronen, T., Koponen, A., Tikkanen, H., & Peltonen, J. E. (2021). Variability in haemoglobin mass response to altitude training camps. *Scandinavian Journal of Medicine & Science in Sports*, **31**(1), 44–51.
- Nybo, L., Rønnestad, B., & Lundby, C. (2024). High or hot-perspectives on altitude camps and heat-acclimation training as preparation for prolonged stage races. *Scandinavian Journal of Medicine & Science in Sports*, **34**(1), e14268.
- Oberholzer, L., Siebenmann, C., Mikkelsen, C. J., Junge, N., Piil, J. F., Morris, N. B., Goetze, J. P., Meinild Lundby, A. K., Nybo, L., & Lundby, C. (2019). Hematological adaptations to prolonged heat acclimation in endurance-trained males. *Frontiers in Physiology*, **10**, 1379.

- Park, H. Y., Kim, S., & Nam, S. S. (2017). Four-week “living high training low” program enhances 3000-m and 5000-m time trials by improving energy metabolism during sub-maximal exercise in athletes. *Journal of Exercise Nutrition & Biochemistry*, **21**(1), 1–6.
- Park, H. Y., Park, W., & Lim, K. (2019). Living high-training low for 21 days enhances exercise economy, hemodynamic function, and exercise performance of competitive runners. *Journal of Sports Science and Medicine*, **18**(3), 427–437.
- Peltonen, J. E., Leppavuori, A., Lehtonen, E., Mikkonen, R. S., Kettunen, O., Nummela, A., Ohtonen, O., Gagnon, D. D., Wehrlin, J. P., Wilber, R. L., & Linnamo, V. (2024). Combined intermittent hypoxic exposure at rest and continuous hypoxic training can maintain elevated haemoglobin mass after a hypoxic camp. *Journal of Applied Physiology*, **137**(2), 409–420.
- Périard, J. D., Eijssvogels, T. M. H., & Daanen, H. A. M. (2021). Exercise under heat stress: Thermoregulation, hydration, performance implications, and mitigation strategies. *Physiological Reviews*, **101**(4), 1873–1979.
- Poffé, C., Robberechts, R., Van Thienen, R., & Hespel, P. (2023). Exogenous ketosis elevates circulating erythropoietin and stimulates muscular angiogenesis during endurance training overload. *The Journal of Physiology*, **601**(12), 2345–2358.
- Prommer, N., Sottas, P. E., Schoch, C., Schumacher, Y. O., & Schmidt, W. (2008). Total haemoglobin mass—A new parameter to detect blood doping. *Medicine & Science in Sports & Exercise*, **40**(12), 2112–2118.
- Robach, P., Gammella, E., Recalcati, S., Girelli, D., Castagna, A., Roustit, M., Lundby, C., Lundby, A. K., Bouzat, P., Vergès, S., Séchaud, G., Banco, P., Uhr, M., Cornu, C., Sallet, P., & Cairo, G. (2021). Induction of erythroferrone in healthy humans by micro-dose recombinant erythropoietin or high-altitude exposure. *Haematologica*, **106**(2), 384–390.
- Robach, P., Hansen, J., Pichon, A., Meinild Lundby, A. K., Dandanell, S., Slettalokken Falch, G., Hammarstrom, D., Pesta, D. H., Siebenmann, C., Keiser, S., Kerivel, P., Whist, J. E., Rønnestad, B. R., & Lundby, C. (2018). Hypobaric live high-train low does not improve aerobic performance more than live low-train low in cross-country skiers. *Scandinavian Journal of Medicine & Science in Sports*, **28**(6), 1636–1652.
- Robach, P., & Lundby, C. (2012). Is live high-train low altitude training relevant for elite athletes with already high total haemoglobin mass? *Scandinavian Journal of Medicine & Science in Sports*, **22**(3), 303–305.
- Robach, P., Thoolen, S., Rommel, S., Recalcati, S., Correnti, M., Girelli, D., Castagna, A., Manceau, M., Meinild Lundby, A. K., Hillig, T., Prchal, J. T., Song, J., Stauffer, E., Connes, P., Goetze, J. P., Vergès, S., Mulder, E., Ulliel-Roche, M., Chabridon, L., ... Lundby, C. (2025). Does one year of continuous hypoxia necessarily increase total haemoglobin mass? (abstract) *23rd International Hypoxia Symposium, Lake Louise, Canada*, 112.
- Robertson, E. Y., Saunders, P. U., Pyne, D. B., Aughey, R. J., Anson, J. M., & Gore, C. J. (2010). Reproducibility of performance changes to simulated live high/train low altitude. *Medicine & Science in Sports & Exercise*, **42**(2), 394–401.
- Rodriguez, F. A., Iglesias, X., Feriche, B., Calderon-Soto, C., Chaverri, D., Wachsmuth, N. B., Schmidt, W., & Levine, B. D. (2015). Altitude training in elite swimmers for sea level performance (Altitude Project). *Medicine & Science in Sports & Exercise*, **47**(9), 1965–1978.
- Ryan, B. J., Goodrich, J. A., Schmidt, W., Kane, L. A., & Byrnes, W. C. (2016). Ten days of intermittent, low-dose carbon monoxide inhalation does not significantly alter haemoglobin mass, aerobic performance predictors, or peak-power exercise tolerance. *International Journal of Sports Medicine*, **37**(11), 884–889.
- Rønnestad, B. R., Hamarsland, H., Hansen, J., Holen, E., Montero, D., Whist, J. E., & Lundby, C. (2021). Five weeks of heat training increases haemoglobin mass in elite cyclists. *Experimental Physiology*, **106**(1), 316–327.
- Rønnestad, B. R., Hansen, J., Bonne, T. C., & Lundby, C. (2022a). Case report: Heat suit training may increase haemoglobin mass in elite athletes. *International Journal of Sports Physiology and Performance*, **17**(1), 115–119.
- Rønnestad, B. R., Lid, O. M., Hansen, J., Hamarsland, H., Mølmen, K. S., Nygaard, H., Ellefsen, S., Hammarström, D., & Lundby, C. (2022b). Heat suit training increases haemoglobin mass in elite cross-country skiers. *Scandinavian Journal of Medicine & Science in Sports*, **32**(7), 1089–1098.
- Rønnestad, B. R., Odden, I., Urianstad, T., Hansen, J., Mølmen, K. S., & Cardinale, D. A. (2025). Heat suit training preserves the increased haemoglobin mass after altitude camp in elite cyclists. *Medicine & Science in Sports & Exercise*, **57**(1), 81–87.
- Rønnestad, B. R., Urianstad, T., Hamarsland, H., Hansen, J., Nygaard, H., Ellefsen, S., Hammarström, D., & Lundby, C. (2022c). Heat training efficiently increases and maintains haemoglobin mass and temperate endurance performance in elite cyclists. *Medicine & Science in Sports & Exercise*, **54**(9), 1515–1526.
- Schmidt, W. F. J., Hoffmeister, T., Haupt, S., Schwenke, D., Wachsmuth, N. B., & Byrnes, W. C. (2020). Chronic exposure to low-dose carbon monoxide alters haemoglobin mass and $\dot{V}O_{2\max}$. *Medicine & Science in Sports & Exercise*, **52**(9), 1879–1887.
- Schmitt, L., Willis, S. J., Fardel, A., Coulmy, N., & Millet, G. P. (2018). Live high-train low guided by daily heart rate variability in elite Nordic-skiers. *European Journal of Applied Physiology*, **118**(2), 419–428.
- Sharma, A. P., Saunders, P. U., Garvican-Lewis, L. A., Périard, J. D., Clark, B., Gore, C. J., Raysmith, B. P., Stanley, J., Robertson, E. Y., & Thompson, K. G. (2018). Training quantification and periodization during live high train high at 2100 M in elite runners: An observational cohort case study. *Journal of Sports Science and Medicine*, **17**(4), 607–616.
- Siebenmann, C., Cathomen, A., Hug, M., Keiser, S., Lundby, A. K., Hilty, M. P., Goetze, J. P., Rasmussen, P., & Lundby, C. (2015). Hemoglobin mass and intravascular volume kinetics during and after exposure to 3,454-m altitude. *Journal of Applied Physiology*, **119**(10), 1194–1201.
- Siebenmann, C., Keiser, S., Robach, P., & Lundby, C. (2017). CORP: Determination of blood volumes in man. *Journal of Applied Physiology*, **123**(3), 645–654.

- Sitkowski, D., Szygula, Z., Surala, O., Orysiak, J., Zdanowicz, R., Pokrywka, A., Starczewski, M., & Malczewska-Lenczowska, J. (2019). Hematological status and endurance performance predictors after low altitude training supported by normobaric hypoxia: A double-blind, placebo controlled study. *Biology of Sport*, **36**(4), 341–349.
- Skattebo, O., & Hallen, J. (2022). Individual variations in pre-altitude haemoglobin mass influence haemoglobin mass responses to repeated altitude sojourns. *Scandinavian journal of Medicine & Science in Sports*, **32**(10), 1493–1501.
- Song, J., Sundar, K., Gangaraju, R., & Prchal, J. T. (2017). Regulation of erythropoiesis after normoxic return from chronic sustained and intermittent hypoxia. *Journal of Applied Physiology* (1985), **123**(6), 1671–1675.
- Stalmans, M., Tominec, D., Lauriks, W., Robberechts, R., Ramaekers, M., Debevec, T., & Poffé, C. (2025). Ketone ester ingestion impairs exercise performance without impacting cognitive function or circulating EPO during acute hypoxic exposure. *Journal of Applied Physiology* (1985), **138**(6), 1309–1320.
- Urianstad, T., Villanova, S., Odden, I., Hansen, J., Molmen, K. S., Porcelli, S., Ronnestad, B. R., & Cardinale, D. A. (2024). Carbon monoxide supplementation: Evaluating its potential to enhance altitude training effects and cycling performance in elite athletes. *Journal of Applied Physiology*, **137**(5), 1092–1105.
- Wehrli, J. P., Marti, B., & Hallén, J. (2016). Hemoglobin mass and aerobic performance at moderate altitude in elite athletes. *Advances in Experimental Medicine and Biology*, **903**, 357–374.
- Yan, B., Ge, X., Yu, J., Hu, Y., & Girard, O. (2021). Hypoxic re-exposure retains haematological but not performance adaptations post-altitude training. *European Journal of Applied Physiology*, **121**(4), 1049–1059.
- Yi, L., Wu, J., Yan, B., Wang, Y., Zou, M., Zhang, Y., Li, F., Qiu, J., & Girard, O. (2024). Effects of three weeks base training at moderate simulated altitude with or without hypoxic residence on exercise capacity and physiological adaptations in well-trained male runners. *PeerJ*, **12**, e17166.

Additional information

Competing interests

Both authors declare that there are no competing interests and that they also have no conflict of interest in accordance with journal policy to disclose.

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Supporting information

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